

Thin-walled shell homogenization of composite beams via the Mechanics of Structure Genome: Kirchhoff and Reissner–Mindlin models with three-dimensional stress recovery, validated against solid finite elements

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Abstract

Slender composite structures—wind-turbine blades, rotor blades and aerospace stiffeners—are routinely reduced to one-dimensional (1D) beams whose cross-sectional constitutive law is obtained by homogenization. For thin-walled sections, a shell idealization of the wall is far cheaper than a full two-dimensional (2D) solid discretization, but it must reproduce the same 6×6 Timoshenko stiffness *and* the same recovered three-dimensional (3D) stress field. We present a thin-walled cross-sectional analysis built on the Mechanics of Structure Genome (MSG), implemented in JAX without a finite-element library or MPI. The wall is modelled by a 1D shell structure genome whose through-thickness behaviour is described by an 8×8 plate constitutive law: a classical-lamination-theory 6×6 membrane–bending block augmented by a 2×2 transverse-shear block obtained from a first-order shear-deformation (FSDT) 3D-equilibrium argument. Two shell kinematics are formulated on the same genome: a Kirchhoff model (C^1 Hermite, no transverse shear) and a Reissner–Mindlin (RM) model (C^0 Lagrange, transverse shear retained). Euler–Bernoulli and Timoshenko stiffnesses follow from the asymptotic fluctuation fields $\mathbf{V}_0$ and $\mathbf{V}_1$, and the same fields drive a two-step dehomogenization that recovers the 3D stress, including a parabolic transverse-shear profile for the RM model. The method is validated on an anisotropic single-ply -45° circular tube against a 3D solid cross-sectional analysis (FSDT

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solid elements, FEniCSx). At the published thickness ratio $h/R = 0.121$ every non-zero Timoshenko 6×6 term agrees with the solid to within 0.7% (including the tension–twist coupling), and the recovered in-plane σ_{22} and σ_{12} on a through-thickness path match to within 1.5%. A thickness sweep ($h/R = 0.121 \rightarrow 0.015$) shows that the shell and solid *Timoshenko* stiffnesses coincide to $< 0.15\%$ at all thicknesses, while the only residual discrepancy—the through-thickness gradient of the axial stress—decays linearly with h/R , confirming it as the expected $O(h/R)$ thick-shell limit of the recovery rather than a modelling error. The RM model additionally recovers the transverse-shear stresses that the Kirchhoff model omits by construction. Applied to a wind-turbine blade station, the homogenized 6×6 matches a VABS solid-element reference to a 5.6% relative norm, and the recovered stress in a thin sandwich panel agrees to within 1.4%.

Keywords: composite beam, cross-sectional analysis, Mechanics of Structure Genome, thin-walled shell, Reissner–Mindlin, stress recovery

1. Introduction

Composite beams are ubiquitous in wind-energy and aerospace structures, where the span greatly exceeds the cross-sectional dimensions. Direct 3D finite-element (FE) analysis of such slender bodies is wasteful; the standard remedy is a two-step procedure: (i) a *cross-sectional analysis* that homogenizes the 3D continuum into a 1D beam with effective stiffness and mass, and (ii) a global 1D beam solution, followed by (iii) a *dehomogenization* that recovers the pointwise 3D stress/strain from the 1D beam response [1, 2]. The Variational Asymptotic Beam Sectional analysis (VABS) [3, 2, 4] and, more recently, the Mechanics of Structure Genome (MSG) [5, 6] provide a rigorous, asymptotically correct basis for this reduction.

For thin-walled sections the wall can be represented either by a 2D solid discretization of the through-thickness domain or by a *shell* idealization of the wall mid-surface. The solid representation is general but expensive; the shell representation reduces the cross-section to a one-dimensional (1D) curve and is therefore an order of magnitude cheaper, at the price of the kinematic assumptions of plate/shell theory. A practical thin-walled tool must nonetheless deliver (a) the full 6×6 Timoshenko stiffness, including anisotropic couplings, and (b) an accurate recovered 3D stress field, since the latter drives strength and fatigue assessment.

Two questions motivate the present work. First, how faithfully does a shell-based MSG cross-sectional analysis reproduce a 3D solid analysis, in both the homogenized stiffness and the recovered stress, and how does that fidelity depend on the wall slenderness h/R ? Second, what is gained by retaining transverse shear through a Reissner–Mindlin (RM) shell kinematics relative to the classical Kirchhoff kinematics?

We address these with a thin-walled MSG cross-sectional analysis implemented in JAX, without a finite-element library or MPI dependency. Our contributions are:

1. a unified 1D shell structure-genome formulation supporting both a Kirchhoff (C^1 Hermite) and a Reissner–Mindlin (C^0 Lagrange) wall kinematics on the same mesh and constitutive data;
2. an 8×8 plate constitutive model combining the classical-lamination-theory (CLT) 6×6 block with a 2×2 transverse-shear block obtained from an FSDT 3D-equilibrium recovery, which reduces to the textbook $\frac{5}{6}Gh$ for an isotropic plate;
3. Euler–Bernoulli and Timoshenko stiffness via the asymptotic fluctuation fields, and a consistent two-step dehomogenization that recovers the 3D stress, including the parabolic transverse-shear profile for the RM model;
4. a verification against a 3D solid cross-sectional analysis on an anisotropic -45° tube, comprising the full 6×6 , the recovered 3D stress, and a thickness-convergence study that isolates the $O(h/R)$ thick-shell limit of the recovery.

2. Theory

2.1. Dimensional reduction by the Mechanics of Structure Genome

Let $\mathbf{z} = (z_1, z_2, z_3)$ denote the 3D material point, with z_1 the beam (span-wise) axis and (z_2, z_3) the cross-sectional coordinates. MSG expresses the 3D displacement field as the sum of a 1D beam kinematics and a small fluctuating warping $\mathbf{w}$ that the analysis determines:

$$\mathbf{u}(\mathbf{z}) = \mathbf{u}^{\text{beam}}(z_1) + \mathbf{w}(z_1; z_2, z_3). \quad (1)$$

Substituting into the 3D strain energy and invoking the variational asymptotic method yields a constrained minimization over the warping at fixed

macroscopic (beam) strain measures $\boldsymbol{\gamma}$. With $\boldsymbol{\varepsilon}$ the 3D strain in Voigt form and $\mathbf{C}$ the material stiffness, the leading-order (classical) functional is

$$\Pi(\boldsymbol{\gamma}) = \min_{\mathbf{w}} \frac{1}{2} \int_{\Omega} \boldsymbol{\varepsilon}^T \mathbf{C} \boldsymbol{\varepsilon} d\Omega, \quad \boldsymbol{\varepsilon} = \boldsymbol{\Gamma}_e \boldsymbol{\gamma} + \boldsymbol{\Gamma}_h \mathbf{w}, \quad (2)$$

where Ω is the structure genome, $\boldsymbol{\Gamma}_e$ maps the beam strains to 3D strains and $\boldsymbol{\Gamma}_h$ is the warping strain operator. For the classical (Euler–Bernoulli, EB) theory $\boldsymbol{\gamma} = \{\gamma_{11}, \kappa_1, \kappa_2, \kappa_3\}^T$ (extension, twist, two bending curvatures); the generalized Timoshenko theory adds the two transverse shear measures $\{2\gamma_{12}, 2\gamma_{13}\}$, obtained as a refinement driven by a second warping field $\mathbf{w}^{(1)}$ [2].

2.2. Thin-walled shell structure genome

For a thin-walled section the structure genome degenerates to the wall mid-surface, a 1D closed (or open) curve $s \mapsto (z_2(s), z_3(s))$ equipped with an outward thickness coordinate. The through-thickness behaviour is condensed analytically into a *plate* constitutive law, so the unknown warping lives only on the 1D curve. Each material point carries an orthonormal frame $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$: $\mathbf{e}_1$ is the beam axis, $\mathbf{e}_2$ is tangent to the wall, and $\mathbf{e}_3$ is the through-thickness (stacking) normal directed from the outer mould line (OML) inward (Fig. 1a). The lamina orientation angle is measured in the $\mathbf{e}_1$ – $\mathbf{e}_2$ plane.

2.3. Wall kinematics: Kirchhoff and Reissner–Mindlin

Two shell kinematics are formulated on the same genome.

Kirchhoff model. The wall normal remains normal to the deformed mid-surface; transverse shear strains vanish identically, $2\gamma_{13} = 2\gamma_{23} = 0$. The warping field is the mid-surface displacement $\mathbf{w} = \{w_1, w_2, w_3\}$, and the bending strains involve second tangential derivatives of w , requiring C^1 continuity. We discretize with cubic Hermite elements (six degrees of freedom per node: the three displacements and their tangential derivatives).

Reissner–Mindlin model. The wall normal is allowed to rotate independently, introducing transverse-shear strains and the section rotations (ω_1, ω_2) ; the drilling rotation ω_3 is eliminated by enforcing symmetry of the in-plane shear ($\varepsilon_{12} = \varepsilon_{21}$). The nodal unknowns are $\{w_1, w_2, w_3, \omega_1, \omega_2\}$ and the strain field involves only *first* derivatives, so standard C^0 Lagrange elements suffice. The full thin-walled strain field for both models is given in Section 2.4.

There, the wall twisting curvature carries the macroscopic twist κ_1 with coefficient -1 for the RM model (the remainder carried by the $\dot{\omega}_2$ fluctuation), in contrast to -2 for the Kirchhoff model; using the Kirchhoff value in the RM model corrupts the torsional stiffness GJ .

Because C^0 interpolation of the RM transverse-shear strains is prone to shear locking as $h/R \rightarrow 0$, the transverse-shear energy is integrated with a reduced (one-point) Gauss rule while the membrane-bending energy uses full integration; this selective-reduced-integration cures locking without spurious modes for the present problems.

2.4. Shell-genome geometry and the thin-walled strain field

We now give the thin-walled shell strain field on which both wall models are built, in the prismatic, small-strain form used in the implementation; it follows the MSG shell formulation [5, 6].

Geometry.. The reference line carries an orthonormal beam triad $\mathbf{b}_i$ with $\mathbf{b}_1 = \partial \bar{\mathbf{r}} / \partial x_1$; a wall point is $\mathbf{r}(x_1, y_\eta) = \bar{\mathbf{r}}(x_1) + \epsilon y_\eta \mathbf{b}_\eta$, $\eta \in \{2, 3\}$. The surface base vectors, written in the beam triad, are

$$\hat{\mathbf{a}}_\beta = [\mathbf{b}_1 \ \mathbf{b}_2 \ \mathbf{b}_3] [(e_1 + \tilde{\mathbf{k}}_b \mathbf{x}_b) x_{1;\beta} + \mathbf{x}_{;\beta}^b], \quad \hat{\mathbf{a}}_3 = \hat{\mathbf{a}}_1 \times \hat{\mathbf{a}}_2, \quad (3)$$

with $\tilde{\mathbf{k}}_b$ the skew matrix of the initial curvature $\mathbf{k}_b = [-k_{b2}, k_{b1}, k_{b3}]$; for a prismatic segment this reduces to the cross-sectional contour curvature k_{22} . The shell and beam triads are linked by the direction-cosine matrix $C_{ij}^{ab} = \mathbf{a}_i \cdot \mathbf{b}_j$, with $C_{;\alpha}^{ab} = -\tilde{\mathbf{k}}_\alpha^s C^{ab}$.

Kinematics and drilling elimination.. The 3D displacement is $\mathbf{U} = \mathbf{u} + \epsilon \mathbf{w} - \tilde{\mathbf{x}} \boldsymbol{\theta}$ with warping $\mathbf{w} = [w_1, w_2, w_3]$, beam rotation $\boldsymbol{\theta} = [\theta_1, -u'_3, u'_2]$, and the normalization $\langle\langle w_i \rangle\rangle = 0$. In the RM model the shell rotation is $\boldsymbol{\rho} = C^{ab}(\boldsymbol{\theta} + \boldsymbol{\omega})$; enforcing in-plane shear symmetry $\varepsilon_{12} = \varepsilon_{21}$ fixes the drilling rotation $\phi_3 = \rho_3$ in closed form, eliminating ω_3 and leaving the independent fluctuations $\{w_1, w_2, w_3, \omega_1, \omega_2\}$ (five per node). The Kirchhoff model has no independent rotations—the normal follows the mid-surface slope—leaving $\{w_1, w_2, w_3\}$ with C^1 continuity.

Reissner–Mindlin strain field.. Collecting the plate strains $\Gamma_D = [\varepsilon_{11}, \varepsilon_{22}, 2\varepsilon_{12}, \kappa_{11}, \kappa_{22}, \kappa_{12} + \kappa_{21}]^T$ and the transverse shears $\Gamma_G = [2\gamma_{13}, 2\gamma_{23}]^T$, with $(\dot{\cdot}) = \partial / \partial \zeta_2$ along the con-

tour, $R_n = x_2\dot{x}_3 - x_3\dot{x}_2$, and beam strains $\{\gamma_{11}, \kappa_1, \kappa_2, \kappa_3\}$ (κ_1 the twist),

$$\varepsilon_{11} = \gamma_{11} + x_3\kappa_2 - x_2\kappa_3 + \underline{w'_1}, \quad (4)$$

$$\varepsilon_{22} = \dot{x}_\eta\dot{w}_\eta, \quad 2\varepsilon_{12} = \dot{w}_1 + \kappa_1 R_n + \underline{\dot{x}_\eta w'_\eta}, \quad (5)$$

$$\kappa_{11} = \dot{x}_\eta\kappa_\eta + \omega'_2/\dot{x}_2 + \dots, \quad \kappa_{22} = -\dot{\omega}_1, \quad (6)$$

$$\begin{aligned} \kappa_{12} + \kappa_{21} = & -\kappa_1 + \frac{\dot{\omega}_2}{\dot{x}_2} + \frac{k_{22}}{2} \left(\frac{\dot{x}_3}{\dot{x}_2} \right)^2 (\dot{w}_1 - \kappa_1 R_n) \\ & - k_{22} \frac{\dot{x}_3}{\dot{x}_2^2} \omega_2 - \omega'_1 - \dots, \end{aligned} \quad (7)$$

$$\begin{aligned} 2\gamma_{13} = & \kappa_1 \left[x_2 \left(\dot{x}_2 + \frac{\dot{x}_3^2}{2\dot{x}_2} \right) + \frac{x_3\dot{x}_3}{2} \right] - \frac{\dot{w}_1\dot{x}_3}{2\dot{x}_2} + \frac{\omega_2}{\dot{x}_2} \\ & - \frac{w'_2\dot{x}_3}{2} + w'_3 \left(\dot{x}_2 + \frac{\dot{x}_3^2}{2\dot{x}_2} \right), \end{aligned} \quad (8)$$

$$2\gamma_{23} = \dot{w}_3\dot{x}_2 - \dot{w}_2\dot{x}_3 - \omega_1, \quad (9)$$

where underlined terms are $O(\epsilon^2)$, retained only for the Timoshenko refinement. Two features distinguish the RM field: (i) $\kappa_{22} = -\dot{\omega}_1$ and $\kappa_{12} + \kappa_{21}$ involve only *first* contour derivatives of the fluctuations, so C^0 interpolation suffices; and (ii) the transverse shears $2\gamma_{13}, 2\gamma_{23}$ are retained, carrying ω_1, ω_2 . The coefficient of the macroscopic twist κ_1 in $\kappa_{12} + \kappa_{21}$ is -1 (the remainder carried by $\dot{\omega}_2$), in contrast to -2 for Kirchhoff.

Kirchhoff reduction.. Setting $2\gamma_{13} = 2\gamma_{23} = 0$ and expressing the wall rotation through the mid-surface slope recovers the classical (Kirchhoff) thin-walled field; its curvatures then carry *second* contour derivatives of w , requiring C^1 continuity (cubic Hermite here).

Closed form and the reference surface.. For a symmetric laminate at the center reference, the zeroth-order minimization of Eq. (2) admits the closed form (isotropic case)

$$\begin{aligned} \varepsilon_{11} = \gamma_{11} + x_3\kappa_2 - x_2\kappa_3, \quad \varepsilon_{22} = -\nu\varepsilon_{11}, \quad 2\varepsilon_{12} = 0, \quad \kappa_{11} = \dot{x}_\eta\kappa_\eta, \\ \kappa_{22} = -\nu\dot{x}_\eta\kappa_\eta, \quad \kappa_{12} + \kappa_{21} = -2\kappa_1, \quad 2\gamma_{13} = 2\gamma_{23} = 0, \end{aligned} \quad (10)$$

so for a straight prismatic beam at the center reference the transverse shears vanish and RM reduces to Kirchhoff; at an offset (outer) reference they are

non-zero and the RM energy stays asymptotically correct. The membrane–bending strains then map as $\Gamma_D = \mathbf{B}(s)\{\gamma_{11}, \kappa_1, \kappa_2, \kappa_3\}^\top$, giving the closed-form Euler–Bernoulli stiffness $\mathbf{C}_{EB} = \oint \mathbf{B}(s)^\top \mathbf{D} \mathbf{B}(s) ds$; torsion and the transverse-shear terms require the finite-element fluctuation solve below.

2.5. Plate constitutive law (8×8)

The condensed wall behaviour relates the plate stress resultants $\{\mathbf{N}, \mathbf{M}, \mathbf{Q}\}$ to the plate strains $\{\boldsymbol{\varepsilon}^0, \boldsymbol{\kappa}, \boldsymbol{\gamma}^s\}$ through

$$\begin{Bmatrix} \mathbf{N} \\ \mathbf{M} \\ \mathbf{Q} \end{Bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{B} & \mathbf{0} \\ \mathbf{B} & \mathbf{D} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{G} \end{bmatrix} \begin{Bmatrix} \boldsymbol{\varepsilon}^0 \\ \boldsymbol{\kappa} \\ \boldsymbol{\gamma}^s \end{Bmatrix}. \quad (11)$$

The 6×6 membrane–bending block $[\mathbf{A}, \mathbf{B}; \mathbf{B}, \mathbf{D}]$ is obtained from a through-thickness 1D structure genome (equivalently CLT for prismatic laminates), evaluated about the chosen reference surface; reference shifts use the standard parallel-axis relations so that the geometry and the plate law share the *same* reference and the *same* $\mathbf{e}_3$. The transverse-shear block $\mathbf{G}$ is built from an FSDT 3D-equilibrium argument: the static through-thickness shear-flow shape $g(z) = \int (z - z_{\text{na}}) Q_{11} dz$ defines an effective shear stiffness

$$G_{\text{eff}} = \frac{D_{\text{eff}}^2}{\int \frac{g(z)^2}{G_{13}(z)} dz}, \quad (12)$$

which reduces to the classical $\frac{5}{6} Gh$ for an isotropic plate (our implementation reproduces the $\frac{5}{6}$ factor to 0.1%). The same shear shape $g(z)$ is reused in the dehomogenization (Section 2.7) so that the recovered transverse-shear stress is parabolic and vanishes at the faces.

2.6. Homogenized stiffness

Discretizing Eq. (2) produces the linear system relating the warping degrees of freedom to the beam strains. The classical (EB) fluctuation $\mathbf{V}_0$ solves

$$\mathbf{D}_{hh} \mathbf{V}_0 = -\mathbf{D}_{he}, \quad \mathbf{D}_{\text{eff}} = \mathbf{D}_{ee} + \mathbf{V}_0^\top (-\mathbf{D}_{he}), \quad (13)$$

where $\mathbf{D}_{hh}, \mathbf{D}_{he}, \mathbf{D}_{ee}$ are the warping–warping, warping–strain and strain–strain blocks, and $\mathbf{D}_{\text{eff}}$ is the 4×4 EB stiffness. Rigid-body and macroscopic kernels are removed through the constraint operator conjugate to the kernel

Ψ (the three translations and the cross-sectional twist mode $w_2 = -z_3$, $w_3 = z_2$, $\omega_1 = -1$). The generalized Timoshenko (6×6) stiffness follows from the second-order shear-warping field $\mathbf{V}_1$ and the standard transformation built from $\mathbf{Q} = [0, 0; 0, 0; 0, -1; 1, 0]$ [2], condensing the shear flexibility into the 6×6 matrix $\mathbf{C}_6$ ordered as {extension, shear₂, shear₃, twist, bending₂, bending₃}.

2.7. Dehomogenization (3D stress recovery)

Given the 1D beam response (either an applied resultant $\mathbf{F} = \{F_1, \dots, M_3\}$ or, directly, the beam strain $\boldsymbol{\gamma} = \mathbf{C}_6^{-1} \mathbf{F}$), the macroscopic strain and its span-wise derivatives are reconstructed, the warping fields $\mathbf{V}_0, \mathbf{V}_1$ are superposed, and the recovered plate strains are mapped to the 3D strain through the wall thickness. The in-plane components ($\sigma_{11}, \sigma_{22}, \sigma_{12}$) and the negligible σ_{33} follow from the plane-stress-consistent constitutive law evaluated at depth z . The transverse-shear stresses are recovered from the FSDT shape:

$$\sigma_{13}(z) = \frac{V_{13}}{D_{\text{eff}}} g(z), \quad \sigma_{23}(z) = \frac{V_{23}}{D_{\text{eff}}} g(z), \quad (14)$$

with the section transverse-shear resultant carried by the RM shear-warping field. The Kirchhoff model has no transverse-shear kinematics, so it returns $\sigma_{13} = \sigma_{23} \equiv 0$ identically; the RM model returns the parabolic profile of Eq. (14).

2.8. Assumptions and scope

The formulation assumes (i) a prismatic, linearly elastic cross-section; (ii) a thin wall, so that plate kinematics hold through the thickness and the recovered fields are accurate to $O(h/R)$; (iii) plane-stress through the wall ($\sigma_{33} \approx 0$); and (iv) for the present implementation, a flat plate constitutive law (the $(1+z/R)$ curvature metric of the wall is not included in the through-thickness integration). The consequences of (ii) and (iv) are quantified by the thickness study in Section 4.4.

3. Numerical example: anisotropic circular tube

We adopt the single-ply anisotropic circular tube used as a benchmark in the MSG literature. The geometry is a circular tube of mid-surface radius $R = 0.0715$ m and wall thickness $h = 0.008682$ m, giving a thickness ratio $h/R = 0.121$. The wall is a single -45° lamina with the orthotropic

Table 1: Lamina properties of the anisotropic tube.

E_1 (GPa)	$E_2 = E_3$ (GPa)	$G_{12} = G_{13} = G_{23}$ (GPa)	$\nu_{12} = \nu_{13} = \nu_{23}$	ply angle
37	9	4	0.3	-45°

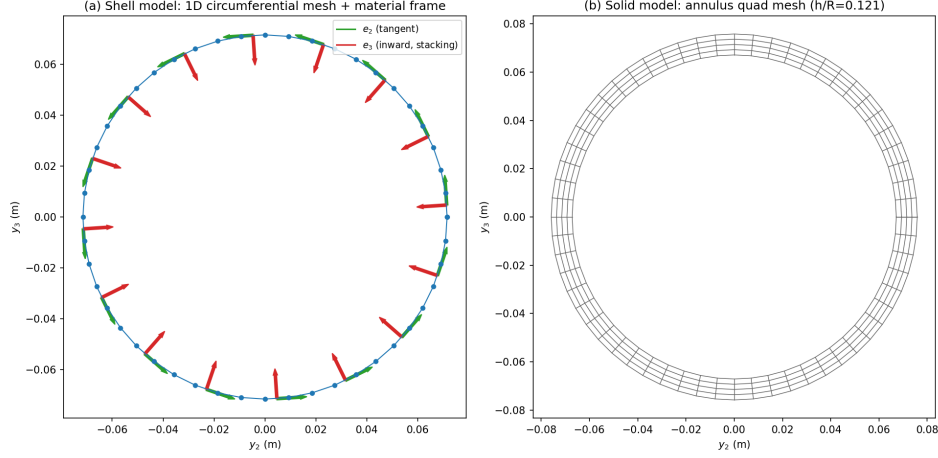


Figure 1: Anisotropic tube: (a) the shell model is a 1D circumferential mesh of the mid-surface, showing the material frame e_2 (tangent) and e_3 (inward, stacking direction); (b) the solid reference model is an annulus of quadrilateral elements ($h/R = 0.121$).

properties of Table 1. The reference surface is the mid-surface (center reference); the shell nodes lie on the outer mould line and are offset by $h/2$ to the mid-surface, with the plate law shifted to the same reference so that geometry and constitutive data are consistent. The published classical stiffness for this case is $C_{11} = 47.785 \times 10^6$, $C_{14} = -0.93755 \times 10^6$ (tension–twist), $C_{44} = 0.14896 \times 10^6$ (GJ) and $C_{55} = C_{66} = 0.10710 \times 10^6$ (EI) (consistent units).

Solid reference.. The 3D solid cross-sectional analysis discretizes the wall annulus with first-order quadrilateral elements (Fig. 1b) and applies the same MSG/VABS reduction with full 3D constitutive behaviour and the lamina orientation baked into the element frame. It is implemented in FEniCSx/dolfinx and serves as the high-fidelity benchmark; its 6×6 was verified to be mesh-converged (invariant to within 0.01% for 6–40 through-thickness elements and 120–360 circumferential elements).

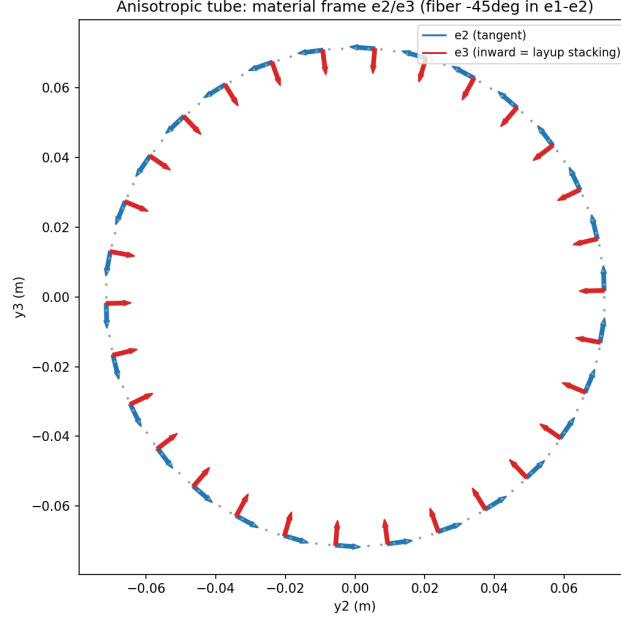


Figure 2: Material frame on the shell mid-surface: $\mathbf{e}_2$ tangent (blue) and $\mathbf{e}_3$ inward (red), consistent around the ring. The -45° fibre lies in the $\mathbf{e}_1$ – $\mathbf{e}_2$ plane. The same $\mathbf{e}_3$ defines both the reference offset and the lamination stacking, an enforced consistency check.

4. Results

4.1. Homogenization

Table 2 compares every non-zero term of the 6×6 Timoshenko stiffness for the Kirchhoff and RM shell models against the solid benchmark and the published classical value. **All shell terms agree with the solid to within 0.7%**, including the tension–twist coupling C_{14} produced by the off-axis ply and the transverse-shear stiffness $C_{22} = C_{33}$. The bending terms $C_{55} = C_{66}$ exceed the published *classical* value by $\sim 15\%$; this is not an error but the difference between the Euler–Bernoulli stiffness (to which the published value belongs, and which our shell EB matches to 0.2%) and the generalized Timoshenko stiffness, whose bending diagonal includes the bending–shear coupling intrinsic to the -45° ply. The solid and both shell models reproduce the *same* Timoshenko value.

Table 2: Anisotropic tube ($h/R = 0.121$, center reference): full Timoshenko 6×6 stiffness. TW-RM and TW-Kirchhoff are the present thin-walled shell models; “solid” is the 3D FE benchmark; the last column is the published classical (EB) value where applicable. Percent errors are relative to the solid Timoshenko benchmark.

term	TW-RM	TW-Kirchhoff	solid	classical	err (RM/Kirch)
C_{11} (EA)	4.7835×10^7	4.7853×10^7	4.7843×10^7	4.7785×10^7	$-0.02\%/+0.02\%$
C_{14} (ext-twist)	-9.3125×10^5	-9.2971×10^5	-9.3459×10^5	-9.3755×10^5	$-0.36\%/-0.52\%$
C_{22} (GA ₂)	1.4502×10^7	1.4450×10^7	1.4548×10^7	—	$-0.31\%/-0.67\%$
C_{33} (GA ₃)	1.4502×10^7	1.4473×10^7	1.4548×10^7	—	$-0.31\%/-0.51\%$
C_{25} (shear-bend)	4.6976×10^5	4.6716×10^5	4.6855×10^5	—	$+0.26\%/-0.30\%$
C_{44} (GJ)	1.4727×10^5	1.4799×10^5	1.4820×10^5	1.4896×10^5	$-0.62\%/-0.14\%$
C_{55} (EI ₂)	1.2266×10^5	1.2243×10^5	1.2281×10^5	1.0710×10^5	$-0.12\%/-0.31\%$
C_{66} (EI ₃)	1.2266×10^5	1.2247×10^5	1.2281×10^5	1.0710×10^5	$-0.12\%/-0.27\%$

4.2. Dehomogenization

A representative beam load $\mathbf{F} = [2 \times 10^4 \text{ N}, 0, 0, 300, 150, 0]$ (axial force, twisting and bending moments) is applied and the 3D stress is recovered along the through-thickness path $z_2 = 0$, $z_3 > 0$ (the top of the tube). Figure 3 overlays the shell and solid recoveries in the material frame. The recovered field satisfies global equilibrium: integrating the recovered σ_{11} over the section returns the applied F_1 , M_2 and M_3 to within 0.2%. The dominant in-plane components agree well: σ_{22} and σ_{12} match the solid to within 1.5%, and the σ_{11} *mean* to within 0.4%. The out-of-plane normal stress σ_{33} is negligible ($\sim 10^{-8}$ of the in-plane level), and the transverse-shear stresses vanish because this load carries no section shear force. The only discernible discrepancy is the through-thickness *gradient* of σ_{11} , analyzed next.

4.3. Isolating the recovery from the stiffness

To separate stiffness differences from recovery differences, both models are driven by the *same* generalized beam strain (pure extension $\varepsilon_{11} = 1$, and pure bending $\kappa_2 = 1$) and the recovered $\sigma_{11}(z)$ is compared (Fig. 4). Under pure extension the through-thickness *means* match to 2.3%, but the small through-thickness gradients have opposite sign (shell +13.5%, solid −9.2% of the mean): the thin-shell and 3D recoveries of the off-axis warping differ at this thickness. The gradient is a higher-order quantity superposed on the dominant uniform field; its behaviour with thickness is examined next.

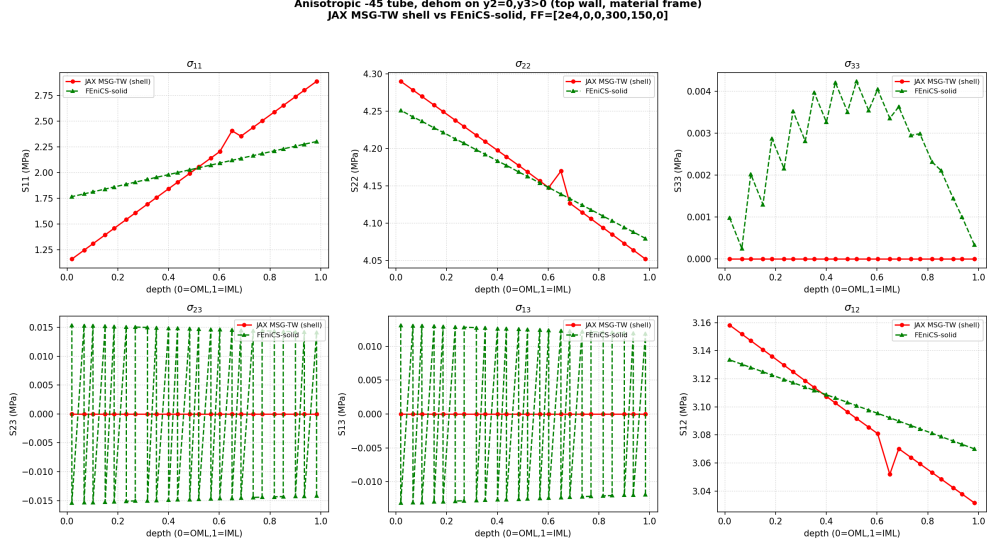


Figure 3: Recovered 3D stress (material frame) on the path $z_2 = 0, z_3 > 0$ under the representative load: shell (MSG-TW) versus solid. In-plane σ_{22} , σ_{12} and the σ_{11} mean agree to within a few percent; $\sigma_{33}, \sigma_{13}, \sigma_{23} \approx 0$ (no section shear for this load).

4.4. Thickness convergence

The wall thickness is reduced at fixed radius, $h/R = 0.121 \rightarrow 0.015$, and the homogenization and strain-driven recovery are repeated for both models and the solid (Fig. 5, Table 3). Two clear conclusions emerge. First, the shell and solid *Timoshenko EI* coincide to within 0.15% at *every* thickness; the ratio of the classical (EB) to the Timoshenko stiffness stays near 0.877, confirming that the $\sim 15\%$ offset relative to the published value is the (thickness-independent) EB-to-Timoshenko difference of the anisotropic section, not a discretization error. Second, the through-thickness σ_{11} gradient—the lone recovery discrepancy—decays linearly with h/R for both models, and the shell–solid gap halves with each halving of h/R ($22.7\% \rightarrow 11.8\% \rightarrow 6.8\% \rightarrow 3.0\%$ of the mean). This is the signature of an $O(h/R)$ thick-shell effect: as the wall becomes thin the shell recovery converges to the 3D solid, as it must. The residual at the benchmark $h/R = 0.121$ is therefore the expected finite-thickness limit of classical shell theory rather than a modelling defect; it can be reduced by a curvature-corrected or higher-order through-thickness recovery.

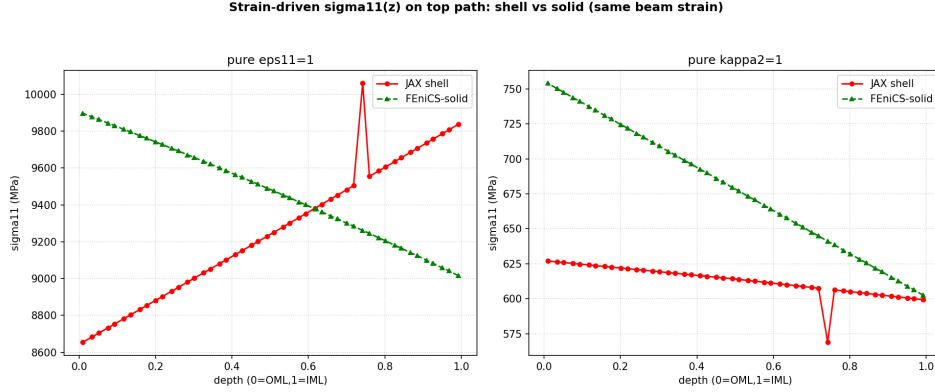


Figure 4: Strain-driven $\sigma_{11}(z)$ on the top path: shell versus solid under identical beam strain (left: pure extension; right: pure bending). The means agree; the through-thickness gradient differs, indicating a recovery (not a stiffness) effect.

Table 3: Thickness convergence. Columns: shell-to-solid Timoshenko EI ratio; through-thickness σ_{11} gradient (% of mean) under pure extension for shell and solid.

h/R	shell _{Timo} /solid _{Timo} EI	shell grad/mean	solid grad/mean
0.121	0.9985	+13.5%	−9.2%
0.061	0.9990	+7.2%	−4.6%
0.030	0.9999	+4.5%	−2.3%
0.015	1.0001	+1.9%	−1.1%

4.5. Out-of-plane stress: Kirchhoff versus Reissner–Mindlin

To exercise the transverse shear, a section shear force $F_2 = 2 \times 10^4$ N (whose shear flow peaks at the top of the tube) is applied and the out-of-plane stresses are recovered on the same path (Fig. 6). The normal stress σ_{33} remains negligible for both shell models and for the solid, consistent with the plane-stress assumption. The transverse-shear stresses σ_{13}, σ_{23} are *identically zero* for the Kirchhoff model (it has no transverse-shear kinematics), whereas the RM model recovers the parabolic through-thickness profile of Eq. (14) that follows the solid. For this thin tube the transverse-shear magnitude is small ($O(h/R)$ below the in-plane shear flow), but the qualitative distinction is exact and becomes important for shear-dominated or thicker sections (e.g. shear webs), where the RM model captures stresses the Kirchhoff model

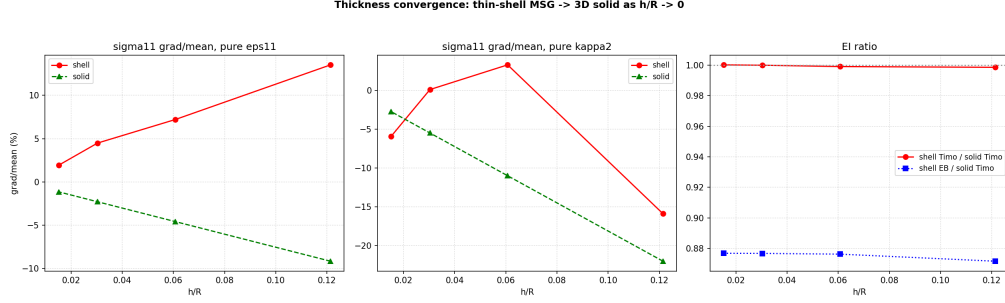


Figure 5: Thickness convergence ($h/R \rightarrow 0$). Left/center: through-thickness σ_{11} gradient (% of mean) under pure extension and pure bending, shell versus solid; both tend to zero linearly in h/R . Right: EI ratios—shell Timoshenko/solid Timoshenko ≈ 1 at all thicknesses, while shell EB/solid Timoshenko stays at ≈ 0.877 .

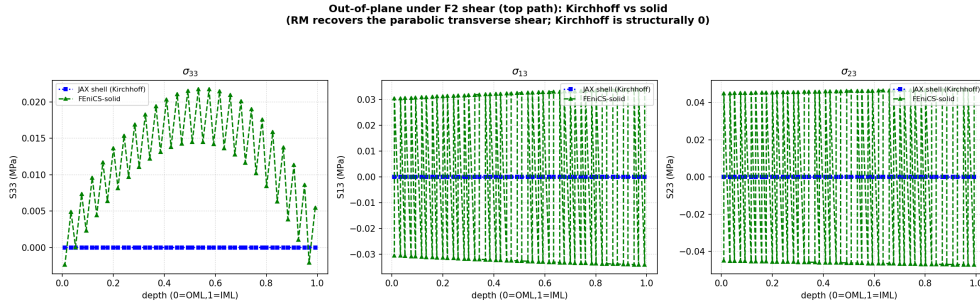


Figure 6: Out-of-plane stress under a section shear force F_2 : σ_{33} , σ_{13} , σ_{23} on the top path. Kirchhoff returns zero transverse shear by construction; the RM recovery follows the solid.

cannot.

5. Application to a wind-turbine blade cross-section

To demonstrate the method on a realistic multi-material section, we analyze station 15 of a utility-scale wind-turbine blade (the `bar_urc` model). The cross-section (Fig. 7a) is a deep, multi-region laminate comprising unidirectional spar caps, two shear webs, foam-cored sandwich panels, and leading-/trailing-edge reinforcements. The high-fidelity reference is VABS, which performs a two-dimensional *solid* finite-element cross-sectional analysis; we use its Timoshenko stiffness (`.K`) and recovered stress (`.SM`) outputs.

Table 4: Station-15 principal Timoshenko stiffnesses: present shell models versus VABS (2D solid FE). The relative norm over all twelve non-zero 6×6 terms is 5.6% for both models.

term	TW-RM	TW-Kirchhoff	VABS	err (RM/Kirch)
C_{11} (EA)	1.3326×10^{10}	1.3326×10^{10}	1.3083×10^{10}	+1.9%/+1.9%
C_{16} (ext-bend)	-3.241×10^9	-3.242×10^9	-3.571×10^9	-9.2%/-9.2%
C_{22} (GA_2)	4.409×10^8	4.462×10^8	4.580×10^8	-3.7%/-2.6%
C_{33} (GA_3)	9.507×10^7	9.627×10^7	1.055×10^8	-9.9%/-8.8%
C_{44} (GJ)	1.595×10^8	1.504×10^8	1.560×10^8	+2.2%/-3.6%
C_{55} (EI_2)	1.6715×10^9	1.6716×10^9	1.6630×10^9	+0.5%/+0.5%
C_{66} (EI_3)	5.809×10^9	5.809×10^9	5.107×10^9	+13.8%/+13.7%

5.1. Homogenization

Table 4 compares the principal Timoshenko stiffnesses of the present shell models against VABS. The axial, shear and flapwise-bending stiffnesses (EA , GA_2 , GA_3 , EI_2) agree to within ~ 2 –10%, and the extension–bending coupling C_{16} to within 9%; the relative norm over all twelve non-zero terms is 5.6% for both shell models. The RM model improves the torsional stiffness (GJ to +2.2% versus -3.6% for Kirchhoff), consistent with its retention of transverse shear. The chordwise bending stiffness EI_3 is high by ~ 14 %—the same thick-section signature quantified for the tube in Section 4.4, here because the deep root section is locally thick relative to its radius of curvature.

5.2. Dehomogenization on a thin sandwich panel

Guided by the thickness study (Section 4.4), which shows the shell recovery converging to the solid as the wall thins, we recover the 3D stress on a through-thickness path in the low-pressure fore sandwich panel (Fig. 7). Although the panel is 40 mm overall, its load-carrying laminate is two *thin* glass skins (4 and 2 mm) separated by a soft foam core—the archetypal thin-walled-shell region. Under a representative blade-section load, the recovered material-frame stress (Fig. 8) matches VABS closely: the dominant axial stress σ_{11} , carried almost entirely by the skins, agrees to within 1.4% (21.7 versus 21.4 MPa peak), while the foam core is correctly near stress-free. The in-plane shear σ_{12} tracks VABS to within its small magnitude, and the out-of-plane components σ_{33} , σ_{13} , σ_{23} are negligible, as expected away from junctions. The Kirchhoff and RM in-plane recoveries coincide (they share the

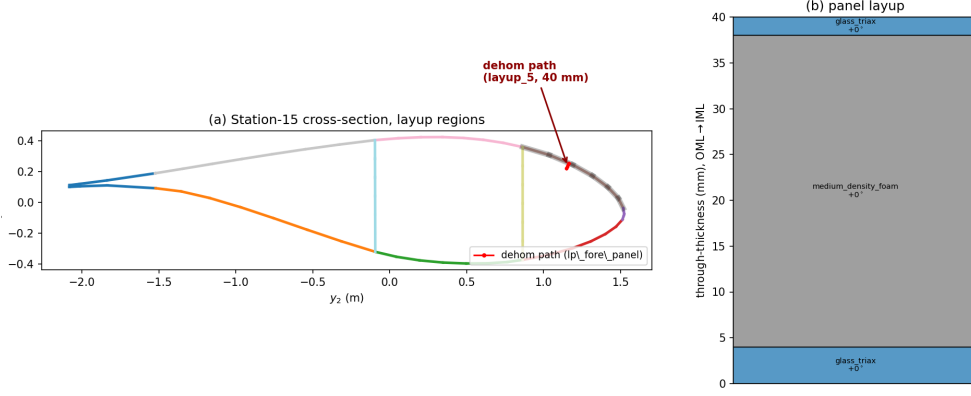


Figure 7: (a) Station-15 cross-section with layup regions (spar caps, two shear webs, sandwich panels, LE/TE reinforcements); the highlighted line is the through-thickness dehomogenization path in the low-pressure fore sandwich panel. (b) The panel layup: thin glass-triax structural skins (4 and 2 mm) bonding a 34 mm foam core.

membrane-bending response); the RM model additionally returns the small parabolic transverse-shear profile that Kirchhoff sets to zero. This confirms, on a realistic blade laminate, that in the thin-walled regions for which the model is intended the shell dehomogenization reproduces the solid (VABS) stress field.

6. Discussion

The results establish that a 1D thin-walled MSG shell analysis reproduces a 3D solid cross-sectional analysis to engineering accuracy for the anisotropic benchmark. Three points merit emphasis. (1) The apparent $\sim 15\%$ disagreement of the bending stiffness with the published value is a comparison artefact: the published number is the classical (EB) stiffness, whereas the relevant quantity for a deformable beam is the generalized Timoshenko stiffness; shell and solid agree on the latter to 0.15% at all thicknesses, and the EB-to-Timoshenko gap is an intrinsic, thickness-independent property of the off-axis section. (2) The dominant recovered stresses (σ_{22} , σ_{12} , the σ_{11} mean) match the solid to within a few percent; the only residual is the through-thickness σ_{11} gradient, which the thickness study identifies unam-

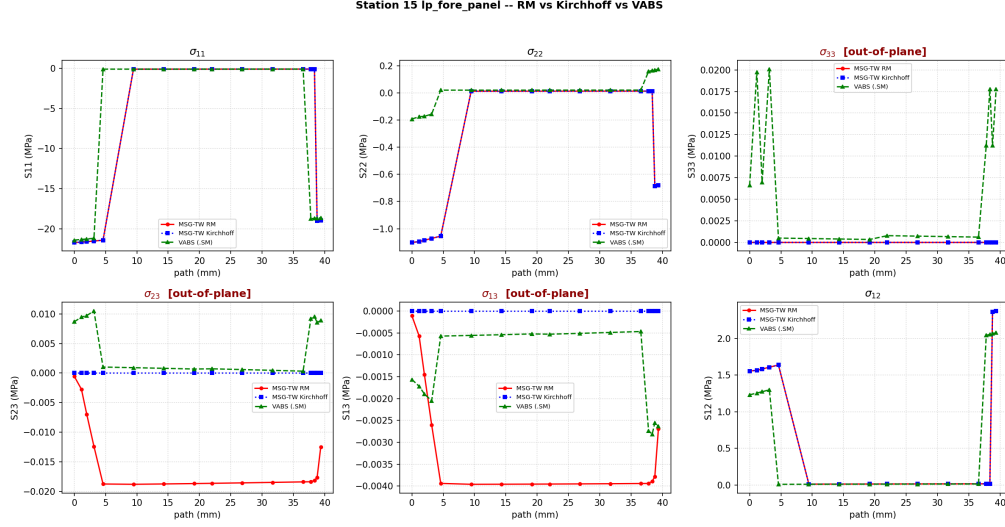


Figure 8: Station-15 recovered 3D stress (material frame) through the thickness of the low-pressure fore sandwich panel: present shell models (TW-RM, TW-Kirchhoff) versus VABS. The axial stress σ_{11} concentrates in the thin skins and matches VABS within 1.4%; the foam core is near stress-free; out-of-plane components are negligible.

biguously as an $O(h/R)$ thick-shell recovery effect that vanishes in the thin limit. For thick walls ($h/R \gtrsim 0.1$) this gradient should be recovered with a curvature-corrected plate metric or a higher-order through-thickness theory. (3) Retaining transverse shear (RM) is essential when the recovered transverse-shear stress is of interest—a quantity that the Kirchhoff model omits identically—at negligible additional cost (C^0 Lagrange versus C^1 Hermite), provided shear locking is controlled by selective reduced integration.

The thin-walled shell analysis reduces the cross-section to a 1D problem and is implemented without a finite-element library or MPI, making it attractive for design-loop and optimization use where many cross-sections must be evaluated; the 3D solid analysis remains the reference for thick or geometrically complex sections.

7. Conclusions

A thin-walled cross-sectional analysis for composite beams was formulated on a 1D shell structure genome with two interchangeable wall kine-

matics (Kirchhoff C^1 and Reissner–Mindlin C^0), an 8×8 plate constitutive law combining CLT with an FSDT transverse-shear block, asymptotic Euler–Bernoulli/Timoshenko stiffness, and a consistent 3D stress recovery. Validated against a 3D solid analysis on an anisotropic -45° tube:

- every Timoshenko 6×6 term matches the solid to within 0.7% at $h/R = 0.121$, including the tension–twist coupling;
- the recovered in-plane σ_{22}, σ_{12} and the σ_{11} mean match to within 1.5%, with global equilibrium satisfied to 0.2%;
- a thickness sweep shows the shell and solid Timoshenko stiffnesses coincide to $< 0.15\%$ at all h/R , while the sole recovery discrepancy (the σ_{11} gradient) decays linearly with h/R , confirming an $O(h/R)$ thick-shell limit;
- the RM model recovers the transverse-shear stresses that the Kirchhoff model omits by construction;
- on a realistic wind-turbine blade station (multi-region laminate with spar caps, shear webs and sandwich panels) the homogenized 6×6 matches a VABS solid-element reference to a 5.6% relative norm, and the recovered stress in a thin sandwich panel agrees with VABS to within 1.4%.

Future work will incorporate the wall-curvature metric in the through-thickness recovery to extend the accuracy of the σ_{11} gradient into the thick-wall regime, and will apply the method to additional multi-cell airfoil sections and junction regions.

Data and code availability

The cross-sectional analysis and the scripts that generate every figure and table are available from the author on reasonable request.

Declaration of competing interest

The author declares no competing interests.

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